

Boss Challenge — Paper 25 (Class 7)

Forests | Wastewater Story | The Triangle & its Properties | Exponents & Powers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Why do people describe a thick forest as the 'green lungs' of our planet? Because trees:
(A) release carbon dioxide and take in oxygen
(B) give out only water vapour and no gases
(C) store all the rainwater inside their leaves
(D) release oxygen and take in carbon dioxide
2. A forester leans a 5 m ladder against a tree so its foot is 3 m from the base of the trunk. By the right-angle (Pythagoras) property, how high up the trunk does the ladder reach?
(A) 8 m
(B) 6 m
(C) 4 m
(D) 2 m
3. The used, dirty water from homes, toilets and factories is called:
(A) fresh water
(B) distilled water
(C) rainwater
(D) wastewater (sewage)
4. One bacterium in untreated sewage doubles every hour: 1 → 2 → 4 → 8 ... After 5 hours the number is 2ⁿ, which equals:
(A) 32
(B) 10
(C) 25
(D) 16
5. The animals and microorganisms that break down dead plants and animals in a forest are called:
(A) predators
(B) producers
(C) consumers
(D) decomposers
6. If you measure and add up all three corner angles inside any triangle, the total comes to:
(A) 270°
(B) 360°
(C) 90°
(D) 180°

7. The network of underground pipes that carries sewage away is called the:
- (A) rainwater drain only
 - (B) water supply line
 - (C) sewage
 - (D) sewerage
8. The value of 3^4 is:
- (A) 12
 - (B) 81
 - (C) 64
 - (D) 27
9. The branches of the tall trees in a forest spread out and join to form a roof-like cover called the:
- (A) canopy
 - (B) humus
 - (C) understorey
 - (D) crown
10. A triangle in which all three sides are equal in length is called:
- (A) right-angled
 - (B) isosceles
 - (C) scalene
 - (D) equilateral
11. The harmful and unwanted substances dissolved or floating in wastewater are called:
- (A) gases only
 - (B) contaminants
 - (C) minerals only
 - (D) nutrients only
12. In the expression 5^3 , the number 3 is called the:
- (A) exponent (power)
 - (B) root
 - (C) base
 - (D) product
13. The branchy top part of one single tree is called its:
- (A) crown
 - (B) humus
 - (C) canopy
 - (D) trunk
14. An exterior angle of a triangle is equal to the:
- (A) interior angle right next to it
 - (B) sum of the two interior opposite angles
 - (C) difference of the other two angles
 - (D) sum of all three interior angles

15. The place where wastewater is cleaned before being let out is called a:

- (A) water tank
- (B) wastewater treatment plant
- (C) power plant
- (D) dairy

16. Using the laws of exponents, $a^m \times a^n$ equals:

- (A) $a^{(m \times n)}$
- (B) $a^{(m-n)}$
- (C) $a^{(m/n)}$
- (D) $a^{(m+n)}$

17. The dark-coloured, nutrient-rich layer formed in forest soil by the decay of dead leaves and animals is called:

- (A) humus
- (B) sand
- (C) bedrock
- (D) clay

18. Two angles of a triangular forest plot measure 50° and 60° . The third angle is:

- (A) 90°
- (B) 110°
- (C) 80°
- (D) 70°

19. Large floating objects like rags, sticks and plastic are first removed from sewage by passing it through:

- (A) a boiler
- (B) a sand filter
- (C) bar screens
- (D) an aeration tank

20. Using the laws of exponents, a^m / a^n (with $a \neq 0$) equals:

- (A) $a^{(m \times n)}$
- (B) $a^{(m-n)}$
- (C) $a^{(n-m)}$
- (D) $a^{(m+n)}$

21. Forests help prevent floods mainly because:

- (A) they make the rain stop falling
- (B) their roots and dead leaves let rainwater soak slowly into the ground
- (C) their leaves drink up all the river water
- (D) they heat the air so the clouds disappear

22. A triangle that has one angle exactly equal to 90° is called a:
- (A) right-angled triangle
 - (B) acute-angled triangle
 - (C) obtuse-angled triangle
 - (D) equilateral triangle
23. After the bar screens, sand, grit and small pebbles are allowed to settle out in the:
- (A) grit and sand removal tank
 - (B) bar screen
 - (C) chlorine tank
 - (D) aeration tank
24. Using the laws of exponents, $(a^m)^n$ equals:
- (A) $a^{(m-n)}$
 - (B) $a^{(mxn)}$
 - (C) $a^{(m+n)}$
 - (D) a
25. A forest is described as a 'dynamic living entity' because:
- (A) it can move from place to place
 - (B) it never changes at all
 - (C) it is full of life and keeps changing and renewing itself
 - (D) it is made only of non-living rock
26. A triangle with exactly two equal sides is called:
- (A) isosceles
 - (B) right-angled
 - (C) equilateral
 - (D) scalene
27. The heavy solid waste that sinks to the bottom of a settling tank is called:
- (A) clear water
 - (B) foam
 - (C) scum
 - (D) sludge
28. If you take a number that is not zero and raise it to the power zero, you always get:
- (A) 1
 - (B) 0
 - (C) infinity
 - (D) the number itself
29. In a forest food chain, the green plants are the:
- (A) consumers
 - (B) producers
 - (C) decomposers
 - (D) predators

30. Inside a triangle, a straight segment drawn from a corner to the exact middle of the side facing it is a:
- (A) base
 - (B) hypotenuse
 - (C) altitude
 - (D) median
31. Oils, grease and other light matter that float on the top of the settling tank are skimmed off as:
- (A) clarified water
 - (B) sludge
 - (C) sand
 - (D) scum
32. A treatment plant cleans 10■ litres of water a day. Written out in full, 10■ litres is:
- (A) 60 litres
 - (B) 1,00,000 litres
 - (C) 1,00,00,000 litres
 - (D) 10,00,000 litres (ten lakh)
33. Cutting down large numbers of forest trees (deforestation) leads to:
- (A) less carbon dioxide in the air
 - (B) more rainfall everywhere
 - (C) more carbon dioxide staying in the air
 - (D) more oxygen in the air
34. The perpendicular line drawn from a vertex of a triangle to its opposite side is called the:
- (A) hypotenuse
 - (B) median
 - (C) altitude
 - (D) base
35. Air is bubbled through the cleared water in the aeration tank so that helpful bacteria can:
- (A) turn the water into oil
 - (B) make the water dirty again
 - (C) remove all the oxygen from the water
 - (D) feed on the remaining human waste and other matter
36. The standard (scientific) form of 47000 is:
- (A) $4.7 \times 10■$
 - (B) 4.7×10^3
 - (C) 47×10^3
 - (D) $0.47 \times 10■$

37. Forests help the water cycle because they:

- (A) recharge groundwater and release water vapour into the air
- (B) make all the rain fall as snow
- (C) turn rivers into deserts
- (D) stop water from ever evaporating

38. Three trees stand so that two sides of the triangle they make are 6 m and 8 m. For a triangle to be possible, the third side must be less than:

- (A) 2 m
- (B) 48 m
- (C) 14 m
- (D) more than 14 m

39. The mass of helpful bacteria that grows and settles during aeration is called:

- (A) grit
- (B) activated sludge
- (C) chlorine
- (D) scum

40. $10^3 / 10^2$ equals:

- (A) 10^3
- (B) 10^2
- (C) 10^1
- (D) 10^0

41. The chief job of microorganisms that rot dead leaves on the forest floor is to:

- (A) turn the soil into solid rock
- (B) remove all nutrients from the soil
- (C) return nutrients to the soil as humus
- (D) make the soil poisonous

42. The triangle inequality states that the sum of the lengths of any two sides of a triangle is:

- (A) equal to the third side
- (B) less than the third side
- (C) greater than the third side
- (D) equal to 180°

43. Before the cleaned water is released, it is often disinfected by adding:

- (A) oil
- (B) sludge
- (C) sand
- (D) chlorine

44. $2^3 \times 2^2$ equals:

- (A) 2^1
- (B) 4^1
- (C) 2^1
- (D) 2^1

45. Which of these is NOT a natural product we get from forests?

- (A) plastic
- (B) medicines
- (C) paper
- (D) gum

46. A right-angled triangular warning sign in the forest has its two shorter sides (legs) measuring 9 m and 12 m. Its longest side (hypotenuse) is:

- (A) 3 m
- (B) 225 m
- (C) 15 m
- (D) 21 m

47. After it is dried, the sludge from a treatment plant is often used as:

- (A) manure
- (B) drinking water
- (C) window glass
- (D) fuel for cars

48. Harmful bacteria in a polluted pond multiply 10 times each day. After 3 days one bacterium becomes 10^3 , which equals:

- (A) 30
- (B) 300
- (C) 100
- (D) 1000

49. The layer of small shrubs and herbs growing below the canopy is the:

- (A) crown
- (B) canopy
- (C) understorey
- (D) humus

50. In a right-angled triangle, the side directly opposite the right angle is called the:

- (A) altitude
- (B) hypotenuse
- (C) base
- (D) median

51. Pouring used cooking oil down the kitchen drain is harmful because the oil:
- (A) cleans the inside of the pipes
 - (B) makes the water flow faster
 - (C) turns into safe drinking water
 - (D) hardens and blocks the drainage pipes
52. The value of (-1) raised to any odd power, such as $(-1)^n$, is:
- (A) 1
 - (B) -1
 - (C) 7
 - (D) 0
53. Animals help forests grow by dispersing seeds when they:
- (A) eat fruits and drop the seeds far from the parent tree
 - (B) cut the trees down for timber
 - (C) eat all the seeds so none can grow
 - (D) press the seeds deep into solid rock
54. Each interior angle of an equilateral triangle measures:
- (A) 180°
 - (B) 60°
 - (C) 90°
 - (D) 45°
55. Where there is no sewer connection, a low-cost on-site way to treat household sewage is a:
- (A) overhead water tank
 - (B) septic tank
 - (C) treatment plant
 - (D) river reservoir
56. The value of (-1) raised to any even power, such as $(-1)^n$, is:
- (A) 1
 - (B) 8
 - (C) -1
 - (D) 0
57. Forests reduce soil erosion mainly because:
- (A) the leaves push the soil away
 - (B) tree roots bind the soil and hold it in place
 - (C) they dry the soil into loose dust
 - (D) the trees make the soil into rock
58. In an isosceles triangle, the two angles opposite the two equal sides are:
- (A) always different sizes
 - (B) always 90° each
 - (C) each equal to 60°
 - (D) equal to each other

59. Which of these should NOT be thrown into a household drain?

- (A) rinse water from a glass
- (B) used cooking oil and tea leaves
- (C) clean tap water
- (D) a little hand soap while washing

60. Which is greater, 2^2 or 5^2 ?

- (A) 5^2 is greater
- (B) 2^2 is greater
- (C) they cannot be compared
- (D) they are equal

61. The top layer of forest soil is dark and fertile mainly because:

- (A) decaying leaves keep adding humus to it
- (B) animals paint it with mud
- (C) the sunlight burns it black
- (D) rainwater bleaches it dark

62. A triangular leaf has a base of 10 cm and a height of 6 cm. Its area is:

- (A) 16 cm^2
- (B) 60 cm^2
- (C) 30 cm^2
- (D) 32 cm^2

63. Letting untreated sewage flow straight into a river is dangerous because it:

- (A) kills all the germs in the river
- (B) spreads water-borne diseases like cholera and typhoid
- (C) makes the river water safe to drink
- (D) adds extra oxygen to the river

64. A wastewater plant processes 5×10^3 litres in the morning and another 5×10^3 litres in the evening. The total processed is:

- (A) 10^3 litres
- (B) 10^4 litres
- (C) 25×10^1 litres
- (D) 5×10^1 litres

65. Forests can be a renewable resource only if:

- (A) new trees are planted to replace those that are cut
- (B) every tree is cut down at once
- (C) the forest is covered with concrete
- (D) no rain is ever allowed to fall

66. A triangle in which all three angles are less than 90° is called an:

- (A) obtuse-angled triangle
- (B) straight-angled triangle
- (C) right-angled triangle
- (D) acute-angled triangle

67. Diseases such as cholera, typhoid and dysentery usually spread through:

- (A) touching dry sand
- (B) breathing clean mountain air
- (C) drinking water polluted with sewage
- (D) eating fresh washed fruit

68. The number 100000000 (a 1 followed by eight zeros) written as a power of ten is:

- (A) 10^8
- (B) 8^1
- (C) 10^8
- (D) 10^8

69. Forests help regulate the local climate by:

- (A) removing all moisture from the air
- (B) keeping the surroundings cooler and the air more humid
- (C) stopping the wind from ever blowing
- (D) making the area hotter and drier

70. When one of the corner angles inside a triangle is bigger than a right angle, the triangle is called an:

- (A) obtuse-angled triangle
- (B) equilateral triangle
- (C) acute-angled triangle
- (D) right-angled triangle

71. Good sanitation means:

- (A) the safe disposal of human waste and keeping surroundings clean
- (B) using more water for fun
- (C) storing waste openly near homes
- (D) throwing rubbish into rivers

72. Expressed as a power of 2, the number 1024 is:

- (A) 2^8
- (B) 2^{12}
- (C) 2^{10}
- (D) 10^2

73. In the food chain grass → deer → tiger, the deer is a:

- (A) producer
- (B) herbivore and a primary consumer
- (C) top carnivore
- (D) decomposer

74. Two tree-trunks lean so that the gap between them is a right-angled triangle. One acute angle is 35° . The other acute angle is:

- (A) 45°
- (B) 65°
- (C) 145°
- (D) 55°

75. Why should people never defecate in the open?

- (A) it cleans the surroundings
- (B) it makes water safe to drink
- (C) it pollutes the soil and water and spreads disease
- (D) it helps plants grow faster

76. The value of $(2^3)^2$ is:

- (A) 32
- (B) 12
- (C) 64
- (D) 256

77. The continuous give-and-take among trees, animals, soil and water in a forest makes it an:

- (A) factory
- (B) ecosystem
- (C) island
- (D) desert

78. A scalene triangle is one in which:

- (A) all three angles are 60°
- (B) all three sides are equal
- (C) exactly two sides are equal
- (D) all three sides are of different lengths

79. Treated water released back into a river is good because it:

- (A) kills the fish in the river
- (B) lowers the pollution of natural water bodies
- (C) adds more germs to the river
- (D) makes the river overflow at once

80. The value of $7^2 + 3^3$ is:

- (A) 58
- (B) 100
- (C) 70
- (D) 76

81. Cutting too many forests can reduce rainfall in an area because:

- (A) trees push the clouds away
- (B) the soil becomes too wet to rain
- (C) fewer trees release less water vapour into the air
- (D) animals drink up all the clouds

82. An equilateral triangular garden has a perimeter of 18 m. The length of each side is:

- (A) 54 m
- (B) 6 m
- (C) 3 m
- (D) 9 m

83. The main aim of treating wastewater is to:

- (A) colour the water blue
- (B) remove pollutants so the water can be safely returned to nature
- (C) turn the water into ice
- (D) make the water taste sweet

84. Which of these is equal to 1?

- (A) $3^1 = 1$
- (B) $5^0 = 5$
- (C) 9^0
- (D) $0^2 = 2$

85. The layer of dead and decaying leaves and twigs lying on the forest floor is called:

- (A) litter
- (B) crown
- (C) canopy
- (D) humus

86. Counting one from each corner to the midpoint of the opposite side, the total number of medians in a triangle is:

- (A) 2
- (B) 3
- (C) 1
- (D) 6

87. Manholes are placed at intervals along a sewer line mainly to:

- (A) let rainwater into the sewer
- (B) generate electricity for the city
- (C) allow workers to clean and check the pipes
- (D) store clean drinking water

88. Bacteria in dirty water double every hour. Starting from 1, how many hours does it take to reach 64 bacteria? ($64 = 2^{\blacksquare}$)

- (A) 32 hours
- (B) 8 hours
- (C) 6 hours
- (D) 64 hours

89. Forests act as natural air cleaners because they:

- (A) turn clean air into poison gas
- (B) give out smoke and dust
- (C) burn up all the oxygen
- (D) absorb carbon dioxide and trap dust on their leaves

90. A vertical forest watch-pole is 12 m tall. A support wire runs from the top of the pole to a peg in the ground 5 m from its foot. The length of the wire is:

- (A) 17 m
- (B) 169 m
- (C) 7 m
- (D) 13 m

91. The dirty water entering a treatment plant is called the:

- (A) scum
- (B) effluent
- (C) influent
- (D) sludge

92. Written in standard (scientific) form, 6,40,000 is:

- (A) $64 \times 10^{\blacksquare}$
- (B) $0.64 \times 10^{\blacksquare}$
- (C) $6.4 \times 10^{\blacksquare}$
- (D) $6.4 \times 10^{\blacksquare}$

93. One strong reason not to cut down too many trees is that it:

- (A) adds more oxygen to the atmosphere
- (B) makes the air much cleaner
- (C) destroys animal homes and lets carbon dioxide build up
- (D) increases the number of wild animals

94. Is it possible to draw a single triangle that contains two right angles inside it?
- (A) Yes, if the triangle is very large
 - (B) Yes, but only if it is isosceles
 - (C) No, because the three angles would then add up to more than 180°
 - (D) Yes, every triangle has two right angles
95. The treated water that finally leaves a wastewater treatment plant is called the:
- (A) effluent
 - (B) grit
 - (C) sludge
 - (D) influent
96. $3^2 \times 3^2 \times 3^2$ equals:
- (A) 3■
 - (B) 3■
 - (C) 9■
 - (D) 3■
97. Which statement about a forest food web is correct?
- (A) every animal eats only one single food
 - (B) many food chains link together because animals eat more than one kind of food
 - (C) decomposers are never part of a food web
 - (D) a food web has no green plants in it
98. In triangle ABC, angle A = 90° and angle B = 45° . The measure of angle C is:
- (A) 90°
 - (B) 35°
 - (C) 45°
 - (D) 135°
99. Why is it important to dispose of human waste safely rather than dumping it untreated?
- (A) untreated waste cleans the rivers
 - (B) untreated waste makes crops grow huge
 - (C) untreated waste produces fresh oxygen
 - (D) untreated waste contaminates water and soil and causes epidemics
100. A village of 10^3 people each produces about 10^2 litres of wastewater a day. The total daily wastewater is $10^3 \times 10^2 =$
- (A) 10^{23} litres
 - (B) 10■ litres
 - (C) 10^1 litres
 - (D) 10■ litres